

Elasticsearch - Introduction

Single-node Elasticsearch:

Download docker-compose

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ wget https://dst-de.s3.eu-west-3.amazonaws.com/elasticsearch_fr/compose_1_node/docker-compose.yml
--2025-07-21 14:13:11-- https://dst-de.s3.eu-west-3.amazonaws.com/elasticsearch_fr/compose_1_node/docker-compose.yml
Resolving dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)... 52.95.154.98, 3.5.204.162
Connecting to dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)|52.95.154.98|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 586 [binary/octet-stream]
Saving to: 'docker-compose.yml'

docker-compose.yml      100%[=====]           586  --.-KB/s    in 0s

2025-07-21 14:13:11 (13.9 MB/s) - 'docker-compose.yml' saved [586/586]
```

Launch the container

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ docker-compose up -d
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Creating network "elasticsearch_es-net" with driver "bridge"
Pulling elasticsearch (docker.elastic.co/elasticsearch/elasticsearch:8.1.2)...
8.1.2: Pulling from elasticsearch/elasticsearch
73c9cfb291da: Pull complete
ad9077a2356d: Pull complete
9ddaa916cdc2: Pull complete
268f5a7162e1: Pull complete
41a66faadeae: Pull complete
4a2591ec32d3: Pull complete
92f74538581d: Pull complete
22d313f98125: Pull complete
602c7b835388: Pull complete
Digest: sha256:aff26414403c2e970b2813d5bb871797b9c9ccea910db98200d43bb5dbfa0fb1
Status: Downloaded newer image for docker.elastic.co/elasticsearch/elasticsearch:8.1.2
Pulling kibana (docker.elastic.co/kibana/kibana:8.1.2)...
8.1.2: Pulling from kibana/kibana
73c9cfb291da: Already exists
ec7df805a77d: Pull complete
257e83b8c3e9: Pull complete
5066f0f5558b: Pull complete
4d6229f21c42: Pull complete
6406ea349d97: Pull complete
801be768ae38: Pull complete
d49bb5151555: Pull complete
e66dbaf20a9f: Pull complete
6dc2702d18a6: Pull complete
a803018cd360: Pull complete
c395487fb95f: Pull complete
a502d1cfa3ea: Pull complete
Digest: sha256:051556e7a2c6818687350f25ea4e66e51fa4edf951a52883b387139008a04d77
Status: Downloaded newer image for docker.elastic.co/kibana/kibana:8.1.2
Creating es-container ... done
Creating kb-container ... done
```

Check the health of the cluster

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ curl -X GET "http://localhost:9200/_cluster/health?pretty"
curl: (56) Recv failure: Connection reset by peer
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$
```

Opened URL <http://localhost:9200/>

```
{
  "name": "a905d622e316",
  "cluster_name": "docker-cluster",
  "cluster_uuid": "CJdGKXFtVw99R6djdXNkg",
  "version": {
    "number": "8.1.2",
    "build_flavor": "default",
    "build_type": "docker",
    "build_hash": "31df9689e80bad366ac20176aa7f2371ea5eb4c1",
    "build_date": "2022-03-29T21:18:59.991429448Z",
    "build_snapshot": false,
    "lucene_version": "9.0.0",
    "minimum_wire_compatibility_version": "7.17.0",
    "minimum_index_compatibility_version": "7.0.0"
  },
  "tagline": "You Know, for Search"
}
```

Inspect Nodes & Indices

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ curl -X GET "http://localhost:9200/_cat/nodes?pretty"
172.20.0.2 44 93 100 16.04 9.83 6.71 cdfhilmrstw * a905d622e316
```

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ curl -X GET "http://localhost:9200/_cat/indices?pretty"
```

Secure Three-Node Cluster:

Create a production directory and download docker-compose file and environment

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ cd prod/
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ wget https://dst-de.s3.eu-west-3.amazonaws.com/elasticsearch_fr/compose_3_nodes/docker-compose.yml
--2025-07-22 07:59:56-- https://dst-de.s3.eu-west-3.amazonaws.com/elasticsearch_fr/compose_3_nodes/docker-compose.yml
Resolving dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)... 3.5.204.162, 52.95.154.98
Connecting to dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)|3.5.204.162|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 8525 (8.3K) [binary/octet-stream]
Saving to: 'docker-compose.yml'

docker-compose.yml           100%[=====] 8.33K  --.-KB/s  in 0s

2025-07-22 07:59:56 (90.2 MB/s) - 'docker-compose.yml' saved [8525/8525]

ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ wget https://dst-de.s3.eu-west-3.amazonaws.com/elasticsearch_fr/.env
--2025-07-22 08:00:04-- https://dst-de.s3.eu-west-3.amazonaws.com/elasticsearch_fr/.env
Resolving dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)... 52.95.155.94, 16.12.20.46
Connecting to dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)|52.95.155.94|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 746 [binary/octet-stream]
Saving to: '.env'

.env                         100%[=====] 746  --.-KB/s  in 0s

2025-07-22 08:00:04 (17.5 MB/s) - '.env' saved [746/746]
```

Increase memory map for ES

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ sudo sysctl -w vm.max_map_count=262144
vm.max_map_count = 262144
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$
```

Start the containers

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ docker-compose up -d
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Creating network "prod_default" with the default driver
Creating volume "prod_certs" with local driver
Creating volume "prod_esdata01" with local driver
Creating volume "prod_esdata02" with local driver
Creating volume "prod_esdata03" with local driver
Creating volume "prod_kibana01" with local driver
Creating prod_setup_1 ... done
Creating prod_es01_1 ... done
Creating prod_es02_1 ... done
Creating prod_es03_1 ... done
Creating prod_kibana_1 ... done
Creating prod_kibana_1 ... done
```

Enable TLS Encryption:

Copy the SSL certificate

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ docker-compose ps
Name                                Command                                State                                Ports
-----
prod_es01_1                        /bin/tini -- /usr/local/bi ...      Up (healthy)                        0.0.0.0:9200->9200/tcp,:::9200->9200/tcp, 9300/tcp
prod_es02_1                        /bin/tini -- /usr/local/bi ...      Up (healthy)                        9200/tcp, 9300/tcp
prod_es03_1                        /bin/tini -- /usr/local/bi ...      Up (healthy)                        9200/tcp, 9300/tcp
prod_kibana_1                      /bin/tini -- /usr/local/bi ...      Up (health: starting)              0.0.0.0:5601->5601/tcp,:::5601->5601/tcp
prod_setup_1                      /bin/tini -- /usr/local/bi ...      Up (healthy)                        9200/tcp, 9300/tcp
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ docker cp prod_setup_1:/usr/share/elasticsearch/
config/certs/ca/ca.crt ./ca.crt
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$
```

Check the cluster's health

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ curl -X GET --cacert ./ca/ca.crt -u elastic:data
scientest "https://localhost:9200/_cluster/health?pretty"
{
  "error" : {
    "root_cause" : [
      {
        "type" : "security_exception",
        "reason" : "unable to authenticate user [elastic] for REST request [/_cluster/health?pretty]",
        "header" : {
          "WWW-Authenticate" : [
            "Basic realm=\"security\" charset=\"UTF-8\"",
            "Bearer realm=\"security\"",
            "ApiKey"
          ]
        }
      }
    ],
    "type" : "security_exception",
    "reason" : "unable to authenticate user [elastic] for REST request [/_cluster/health?pretty]",
    "header" : {
      "WWW-Authenticate" : [
        "Basic realm=\"security\" charset=\"UTF-8\"",
        "Bearer realm=\"security\"",
        "ApiKey"
      ]
    }
  },
  "status" : 401
}
```

Check the creation of the 3 nodes

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ curl -X GET --cacert ./ca/ca.crt -u elastic:data
scientest "https://localhost:9200/_cat/nodes?pretty"
{
  "error" : {
    "root_cause" : [
      {
        "type" : "security_exception",
        "reason" : "unable to authenticate user [elastic] for REST request [/_cat/nodes?pretty]",
        "header" : {
          "WWW-Authenticate" : [
            "Basic realm=\"security\" charset=\"UTF-8\"",
            "Bearer realm=\"security\"",
            "ApiKey"
          ]
        }
      }
    ],
    "type" : "security_exception",
    "reason" : "unable to authenticate user [elastic] for REST request [/_cat/nodes?pretty]",
    "header" : {
      "WWW-Authenticate" : [
        "Basic realm=\"security\" charset=\"UTF-8\"",
        "Bearer realm=\"security\"",
        "ApiKey"
      ]
    }
  },
  "status" : 401
}
```

Check the indices created by the system

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ curl -X GET --cacert ./ca/ca.crt -u elastic:data
scientest "https://localhost:9200/_cat/indices?pretty"
{
  "error" : {
    "root_cause" : [
      {
        "type" : "security_exception",
        "reason" : "unable to authenticate user [elastic] for REST request [/_cat/indices?pretty]",
        "header" : {
          "WWW-Authenticate" : [
            "Basic realm=\"security\" charset=\"UTF-8\"",
            "Bearer realm=\"security\"",
            "ApiKey"
          ]
        }
      }
    ],
    "type" : "security_exception",
    "reason" : "unable to authenticate user [elastic] for REST request [/_cat/indices?pretty]",
    "header" : {
      "WWW-Authenticate" : [
        "Basic realm=\"security\" charset=\"UTF-8\"",
        "Bearer realm=\"security\"",
        "ApiKey"
      ]
    }
  },
  "status" : 401
}
```

Clean the environment file (remove trailing whitespaces). Strip spaces / tabs at line ends and confirm no tabs remain

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ sed -i 's/[[[:space:]]*$//'.env
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ grep -n '$\t' .env
```

Recreate Elasticsearch nodes so they read corrected environment variables

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ docker-compose up -d --force-recreate es01 es02 es03
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Recreating prod_setup_1 ... done
Recreating prod_es01_1 ... done
Recreating prod_es02_1 ... done
Recreating prod_es03_1 ... done
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ docker-compose ps

```

Name	Command	State	Ports
prod_es01_1	/bin/tini -- /usr/local/bi ...	Up (health: starting)	0.0.0.0:9200->9200/tcp,:::9200->9200/tcp, 9300/tcp
prod_es02_1	/bin/tini -- /usr/local/bi ...	Up (health: starting)	9200/tcp, 9300/tcp
prod_es03_1	/bin/tini -- /usr/local/bi ...	Up (health: starting)	9200/tcp, 9300/tcp
prod_kibana_1	/bin/tini -- /usr/local/bi ...	Exit 137	0.0.0.0:5601->5601/tcp,:::5601->5601/tcp
prod_setup_1	/bin/tini -- /usr/local/bi ...	Up (healthy)	9200/tcp, 9300/tcp

Check on the creation of the cluster again

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ curl --cacert ./ca.crt -u elastic:datascientest \
> "https://localhost:9200/_cluster/health?pretty"
{
  "cluster_name" : "docker-cluster",
  "status" : "green",
  "timed_out" : false,
  "number_of_nodes" : 3,
  "number_of_data_nodes" : 3,
  "active_primary_shards" : 5,
  "active_shards" : 10,
  "relocating_shards" : 0,
  "initializing_shards" : 0,
  "unassigned_shards" : 0,
  "delayed_unassigned_shards" : 0,
  "number_of_pending_tasks" : 0,
  "number_of_in_flight_fetch" : 0,
  "task_max_waiting_in_queue_millis" : 0,
  "active_shards_percent_as_number" : 100.0
}
```

Check on the creation of the 3 nodes

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ curl --cacert ./ca.crt -u elastic:datascientest \
> "https://localhost:9200/_cat/nodes?pretty"
172.19.0.5 60 64 100 5.78 9.17 8.84 cdfhilmrstw - es03
172.19.0.4 58 69 100 5.78 9.17 8.84 cdfhilmrstw * es02
172.19.0.3 45 70 100 5.78 9.17 8.84 cdfhilmrstw - es01
```

Check the indices created by the system

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch/prod$ curl --cacert ./ca.crt -u elastic:datascientest \
> "https://localhost:9200/_cat/indices?pretty"
green open .monitoring-es-7-2025.07.22 zRaXQuzPTSwfWZjcAoIhKw 1 1 3129 568 4.8mb 2.1mb
```